Assignment 1

Q1 & Q2

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A AI Fundamentals: Concepts and Methods Homework Assignment

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Problem 1

Draw the computation graph of the following function:

$$f(x_0, x_1, x_2, x_3) = \frac{e^{x_3}}{(x_0 + 3x_1) \cdot x_2^2}$$

with $x_0 = 1, x_1 = -1, x_2 = 2, x_3 = 16$ as in the lecture notes. Obtain all the intermediate outputs, final f, and gradients with respect to x_0, x_1, x_2, x_3. (Note: Show each output above the edge and each gradient below the edge.) (20 points)

Solution.

1. Computation Graph

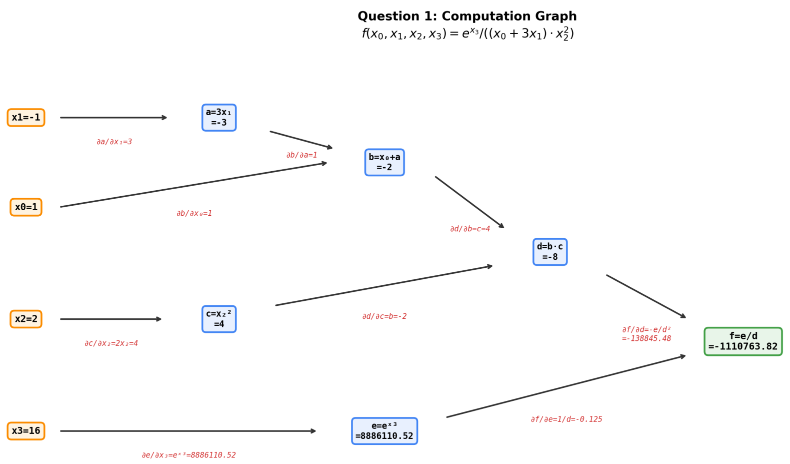

Figure 1: Computation graph of $f(x_0, x_1, x_2, x_3)$ with forward outputs (above edges) and local gradients (below edges in red).

We decompose the function into elementary operations:

- $a = 3 \times x_1$ (Multiplication by constant)
- $b = x_0 + a$ (Addition)
- $c = x_2^2$ (Squaring)
- $d = b \times c$ (Multiplication)
- $e = e^{x_3}$ (Exponential)
- $f = e/d$ (Division)

2. Forward Pass — Intermediate Outputs

Given $x_0 = 1,\ x_1 = -1,\ x_2 = 2,\ x_3 = 16$:

Table 1: Intermediate outputs of the forward pass

Node	Expression	Value
a	$3 \times x_1 = 3 \times (-1)$	-3
b	$x_0 + a = 1 + (-3)$	-2
c	$x_2^2 = 2^2$	4
d	$b \times c = (-2) \times 4$	-8
e	$e^{x_3} = e^{16}$	8,886,110.52
f	$e/d = 8{,}886{,}110.52/(-8)$	$-1{,}110{,}763.82$

3. Backward Pass — Gradients (Chain Rule)

We compute the partial derivatives using the chain rule, starting from the output $\frac{\partial f}{\partial f} = 1$.

Table 2: Gradients computed via the chain rule (backward pass)

Gradient	Derivation	Value
$\frac{\partial f}{\partial f}$	1	1
$\frac{\partial f}{\partial e}$	$\frac{1}{d} = \frac{1}{-8}$	-0.125
$\frac{\partial f}{\partial d}$	$-\frac{e}{d^2} = -\frac{8{,}886{,}110.52}{64}$	$-138{,}845.48$
$\frac{\partial f}{\partial x_3}$	$\frac{\partial f}{\partial e} \cdot e^{x_3} = \frac{e^{x_3}}{d}$	$-1{,}110{,}763.82$
$\frac{\partial f}{\partial b}$	$\frac{\partial f}{\partial d} \cdot c = (-138{,}845.48) \times 4$	$-555{,}381.91$
$\frac{\partial f}{\partial c}$	$\frac{\partial f}{\partial d} \cdot b = (-138{,}845.48) \times (-2)$	277,690.95
$\frac{\partial f}{\partial x_0}$	$\frac{\partial f}{\partial b} \cdot 1$	$\mathbf{-555{,}381.91}$
$\frac{\partial f}{\partial x_1}$	$\frac{\partial f}{\partial b} \cdot 3$	$\mathbf{-1{,}666{,}145.72}$
$\frac{\partial f}{\partial x_2}$	$\frac{\partial f}{\partial c} \cdot 2x_2 = 277{,}690.95 \times 4$	$\mathbf{1{,}110{,}763.82}$

4. Final Gradients Summary

- $\frac{\partial f}{\partial x_0} = -555{,}381.91$
- $\frac{\partial f}{\partial x_1} = -1{,}666{,}145.72$
- $\frac{\partial f}{\partial x_2} = 1{,}110{,}763.82$
- $\frac{\partial f}{\partial x_3} = -1{,}110{,}763.82$

(Verified with PyTorch autograd — results match.)

$\square$

Problem 2

Given the 2-cluster dataset, build an MLP with a single hidden layer of `nbr_hidden_unit` hidden units. Use ReLU as the activation function. Train the MLP with hidden units of 2, 5, 10, 50, 100, 500, 1000 and 5000, and describe what you observe when the number of hidden units varies. (40 points)

Solution.

1. Test Accuracy vs. Number of Hidden Units

Table 3: Test accuracy for different numbers of hidden units

Hidden Units	2	5	10	50	100	500	1000	5000
Accuracy	36.5%	44.5%	62.0%	86.5%	88.0%	88.0%	89.0%	90.5%

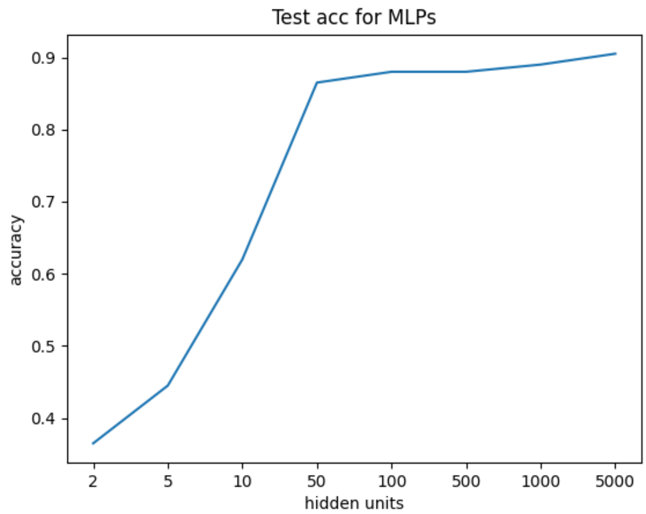

Figure 2: Test accuracy as a function of the number of hidden units.

2. Observations

a) **Accuracy increases with hidden units.** As the number of hidden units increases from 2 to 5000, test accuracy improves significantly—from 36.5% to 90.5%. This demonstrates that higher model capacity enables the MLP to learn more complex decision boundaries.

b) **Rapid improvement in the low range.** The most dramatic accuracy gains occur between 2 and 50 hidden units (36.5% → 86.5%), a nearly 50 percentage point improvement. This indicates that the 2-cluster classification task requires a moderate level of model capacity to separate the two classes.

c) **Diminishing returns at higher capacity.** Beyond 100 hidden units, accuracy improvements become marginal (88.0% → 90.5%). The accuracy curve plateaus, suggesting that additional hidden units provide diminishing returns once the model has sufficient capacity to capture the decision boundary.

d) **Decision boundary complexity.** With very few hidden units (2 or 5), the decision boundaries are nearly linear and cannot separate the two clusters effectively. As hidden units increase (50+), the boundaries become increasingly non-linear and curved, closely conforming to the shape of the data clusters. With 1000 or 5000 units, the boundaries are highly detailed and smooth.

e) **No clear overfitting observed.** Even with 5000 hidden units (far exceeding the data complexity), the test accuracy continues to improve slightly rather than decrease, indicating that the model generalizes well within 30 training epochs. However, with more epochs, overfitting could potentially occur for very large models.

3. Decision Boundary Visualizations

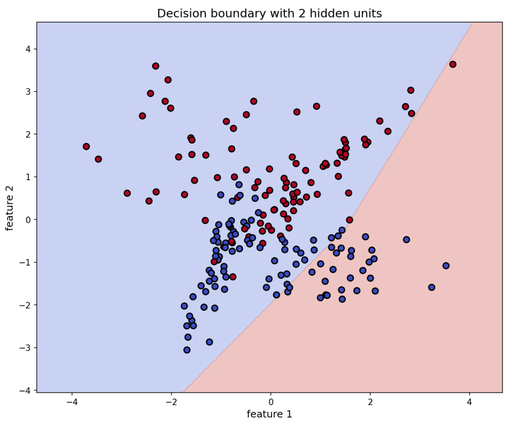

Figure 3: 2 hidden units

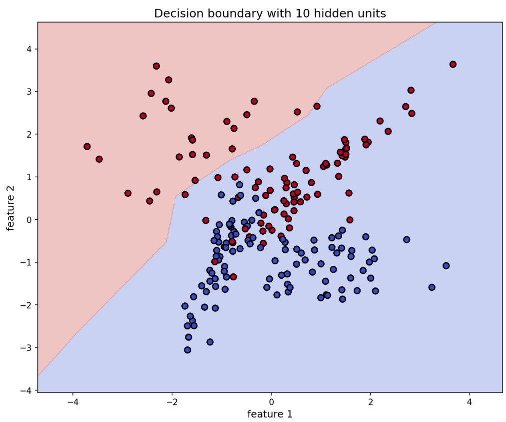

Figure 4: 10 hidden units

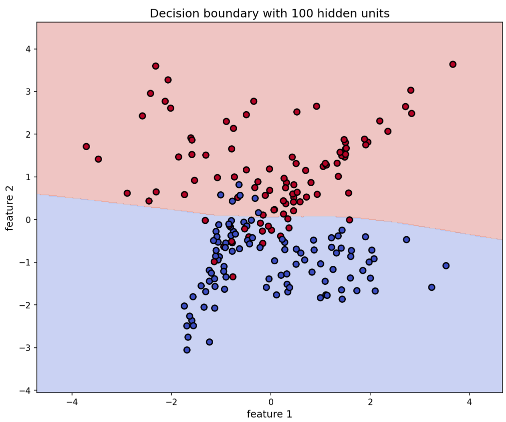

Figure 5: 100 hidden units

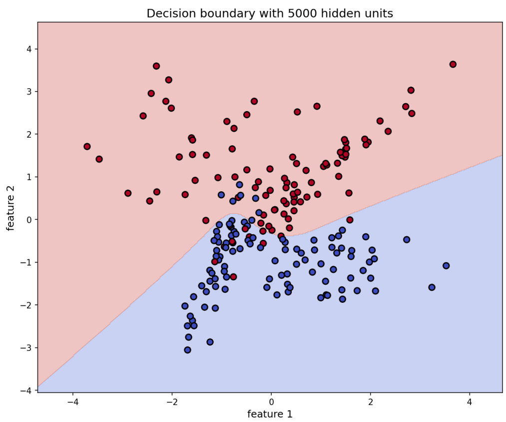

Figure 6: 5000 hidden units

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